

2025 UAA College of Engineering Advisory Board



UAA College of Engineering
UNIVERSITY of ALASKA ANCHORAGE

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Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska



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About Me

Lawrence Giron Jr.

BS Mechanical Engineering – Spring 2023

- ME Research Assistant
- Engineering Academy Teaching Assistant

NASA LaRC Academy – Summer 2023

- Research Associate

NASA GRC Intern – Fall 2023

- Intelligent Control and Autonomy Intern

MS Mechanical Engineering Student – Current

- Projected Graduation – Spring 2025
- Atmospheric Corrosion Research with Dr. Raghu Srinivasan



Introduction

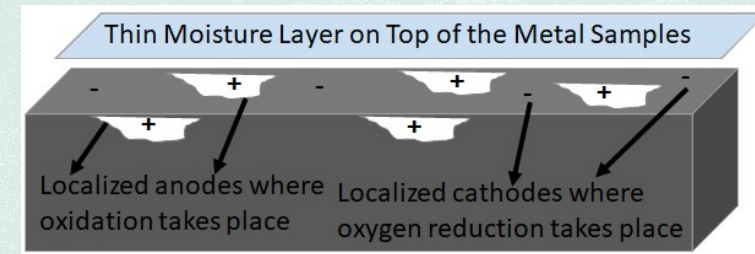
Background

- Comprehensive understanding of atmospheric corrosion allows for accurate estimation of corrosion over time.
- Knowledge gap in the corrosivity of metals in subarctic regions, such as Alaska, where the complexity of corrosion increases.
- One primary factor affecting the rate of atmospheric corrosion is the alternating wet and dry cycles, which involves the time of wetness (TOW) and the formation of thin electrolyte layers.
- ISO 9223, “Corrosion of Metals and Alloys—Corrosivity of Atmospheres—Classification, determination and estimation,” 2012.

r_{corr}	is first-year corrosion rate of metal, expressed in micrometres per year ($\mu\text{m/a}$);
T	is the annual average temperature, expressed in degrees Celsius ($^{\circ}\text{C}$);
RH	is the annual average relative humidity, expressed as a percentage (%);
P_d	is the annual average SO_2 deposition, expressed in milligrams per square metre per day [$\text{mg}/(\text{m}^2\cdot\text{d})$];
S_d	is the annual average Cl^- deposition, expressed in milligrams per square metre per day [$\text{mg}/(\text{m}^2\cdot\text{d})$].

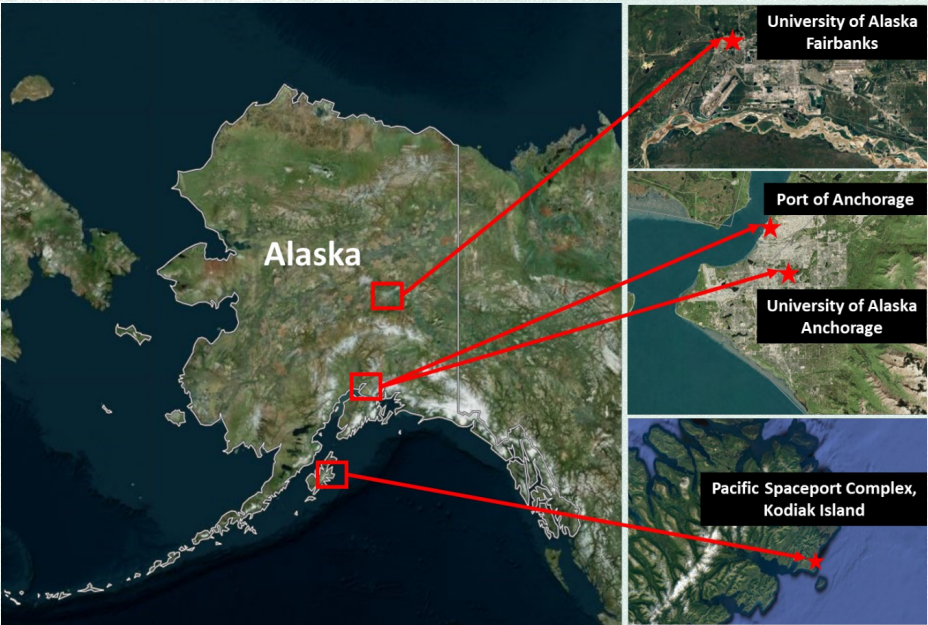
Objectives

- To assemble and deploy multi-angle corrosion test racks with auxiliary weather stations at four test sites across Alaska, USA.
- To monitor and measure the air temperature, relative humidity, aerosol chloride and sulfate deposition, time of wetness at exposure angles of 0° , 30° , and 45° , and rainfall at the test sites across 6-month, 12-month, and 24-month exposure periods.
- To expose five different metal alloys to atmospheric conditions in Alaska at 0° , 30° , and 45° exposure angles.
- To correlate the corrosion rate of metal alloys to exposure conditions and weather parameters to determine the corrosivity category and compare to recognized standards.
- To develop equations to more accurately model and estimate the corrosion rates and evaluate the corrosivity of the atmosphere in Alaska.



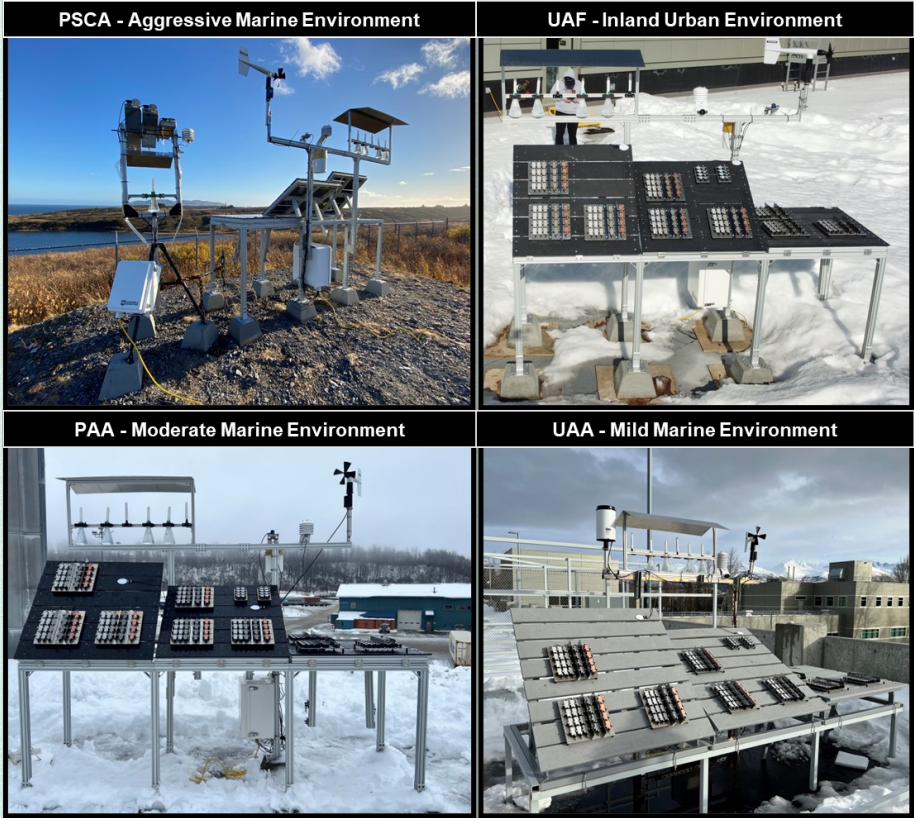
Experimental Method

Test Rack, Weather Station, and Test Sites



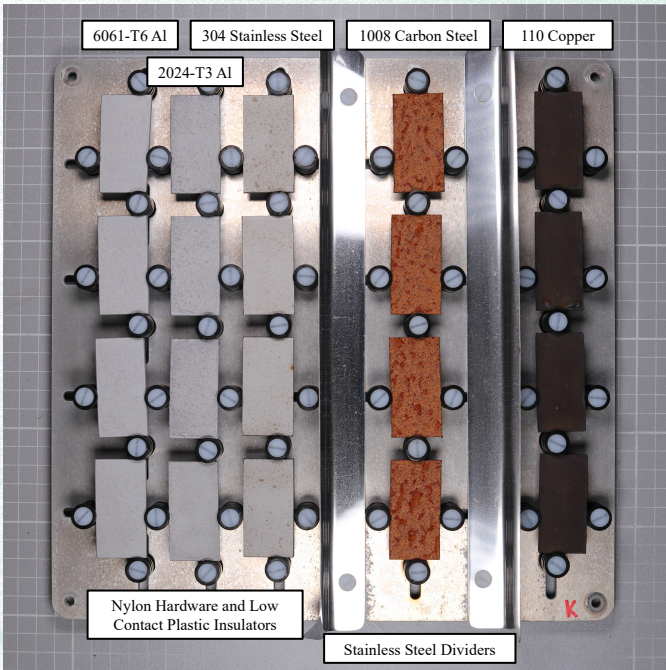
Weather Parameters Measured
Air Temperature [°C]
Relative Humidity [%]
Aerosol Chlorides & Sulfates [mg/(m ² ×day)]
Time of Wetness [mV]
Rainfall [mm]

Location	Coordinates	Environment
Pacific Spaceport Complex – Alaska [Kodiak, AK]	57.453°N, 152.346°W	Aggressive Marine Environment
Port of Alaska [Anchorage, AK]	61.239°N, 149.888°W	Moderate Marine Environment
University of Alaska Fairbanks [Fairbanks, AK]	64.855°N, 147.817°W	Far-inland Urban Environment
University of Alaska Anchorage [Anchorage, AK]	61.192°N, 149.822°W	Inland Mild-Marine Environment



Experimental Method

Metal Alloys and Exposure Periods



Metal Alloys
6061-T6 Aluminum Alloy UNS G96061
2024-T3 Aluminum Alloy UNS A92024
1008 Carbon Steel UNS G10080
304 Stainless Steel UNS G30400
110 Copper UNS C11000

Exposure Periods	
6 Month Set 1	Feb 2022– Aug 2022
6 Month Set 2	Aug 2022– Feb 2022
12 Month Set 1	Feb 2022– Feb 2023
12 Month Set 2	Feb 2023– Jan/Feb 2024
24 Month Set 1	Feb 2022– Jan/Feb 2024

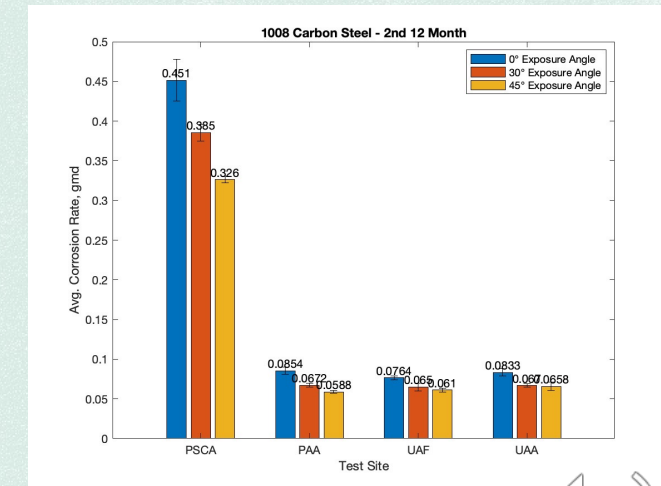
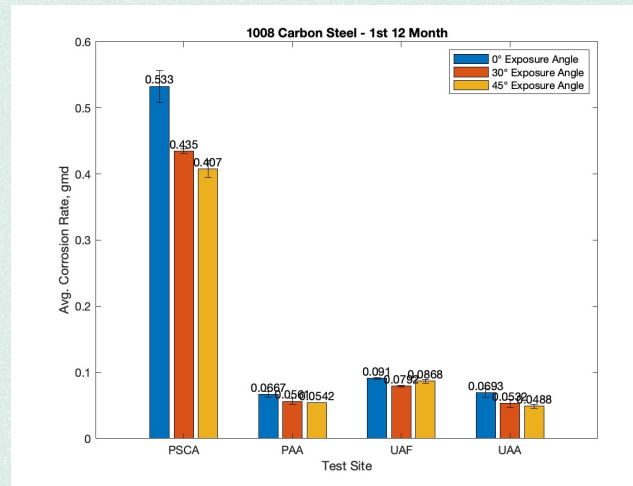
Total Number of Exposed
Coupons in Alaska:

- 4 Locations *
- 5 Metal Alloys *
- 4 Coupons Each *
- 5 Exposure Periods *
- 3 Exposure Angles

= 1,200 Coupons

Corrosion Rate

Designation	Material	Solution	Time	Temperature	Remarks
C.1.2	Aluminum and Aluminum Alloys	1000 mL nitric acid (HNO ₃ , sp gr 1.42)	1 to 5 min	20 to 25 C	Remove extraneous deposits and bulky corrosion products to avoid reactions that may result in excessive removal of base metal
C.2.3	Copper and Copper Alloys	100 mL sulfuric acid (H ₂ SO ₄ , sp gr 1.84) Reagent water to make 1000 mL	1 to 3 min	20 to 25 C	Remove bulky corrosion products before treatment to minimize copper redposition on the specimen surface.
C.3.5	Iron and Steel	500 mL hydrochloric acid (HCl, sp gr 1.19) 3.5 g hexamethylene tetramine . Reagent water to make 1000 mL	10 min	20 to 25 C	Longer times may be required in certain instances.
C.7.1	Stainless Steels	100 mL nitric acid (HNO ₃ , sp gr 1.42) Reagent water to make 1000 mL	20 min	60 C	-

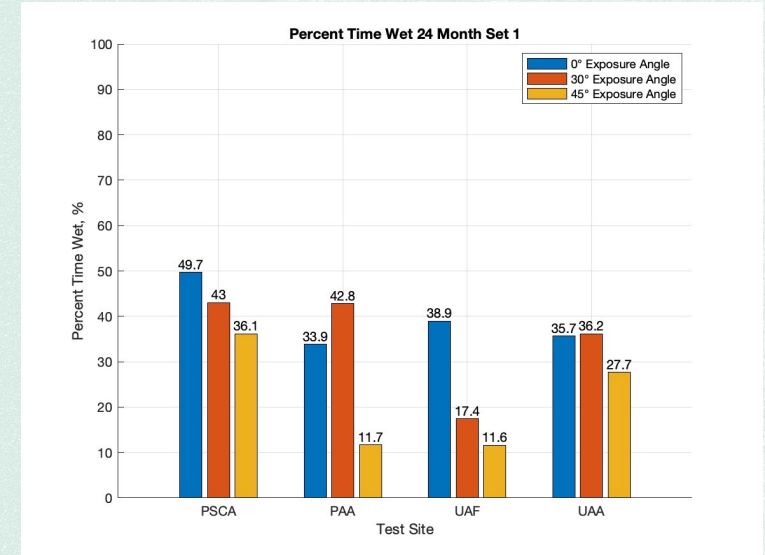
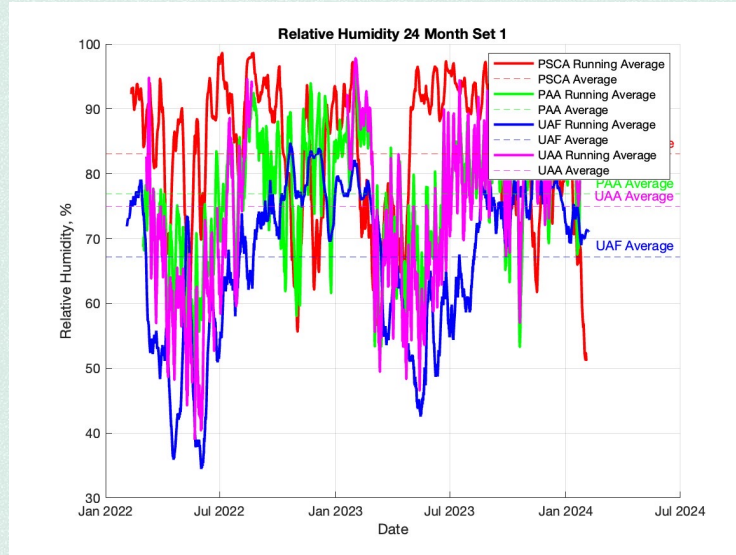
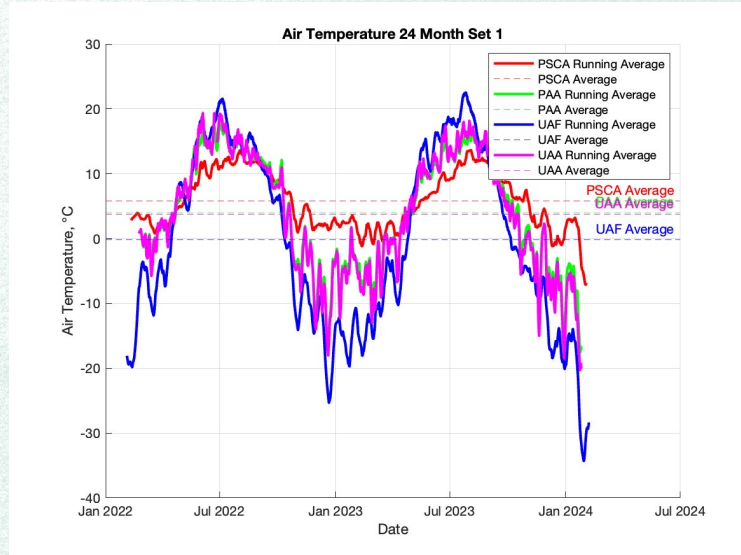


$$\text{Corrosion Rate} = \frac{M_{\text{initial}} - M_{\text{final}}}{SA \cdot D}$$

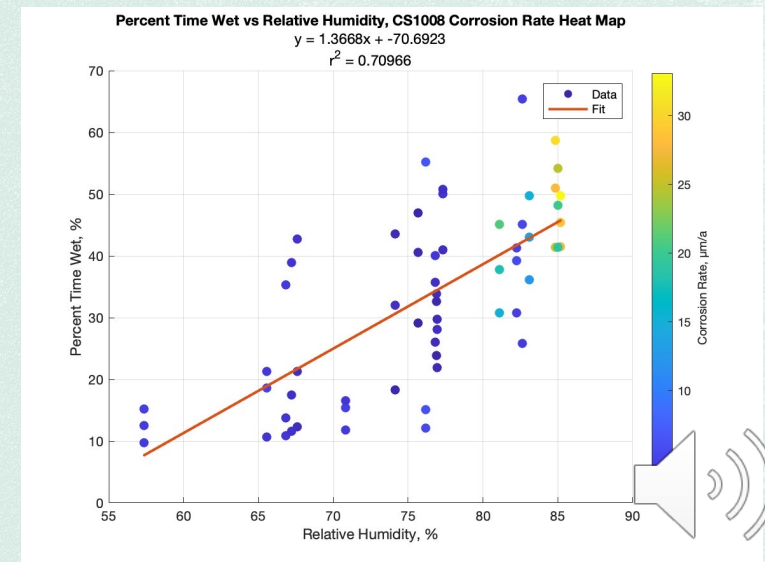


Results and Discussion

Weather Data and Correlations



Location	Exposure Period [Days]	Average Air Temperature [°C]	Average Relative Humidity [%]	Chloride Deposition [mg/(m ² ×day)]	Sulfate Deposition [mg/(m ² ×day)]	Time of Wetness [%]		
						0°	30°	45°
Pacific Spaceport Complex – Alaska [Kodiak, AK]	365	6.26	85.01	47.73	14.99	54.20	48.20	41.41
	357	5.26	81.11	42.78	12.56	45.12	37.74	30.74
Port of Alaska [Anchorage, AK]	370	4.22	76.93	11.33	15.67	28.09	29.75	21.93
	337	3.71	76.82	10.66	12.95	40.08	35.76	25.98
University of Alaska Fairbanks [Fairbanks, AK]	376	-0.07	66.83	9.26	22.89	28.09	29.75	21.93
	357	-0.25	67.58	10.76	19.14	40.08	35.76	25.98
University of Alaska Anchorage [Anchorage, AK]	367	4.05	74.16	8.85	15.09	31.96	43.52	18.29
	335	3.36	77.31	10.49	9.82	50.02	50.78	40.98



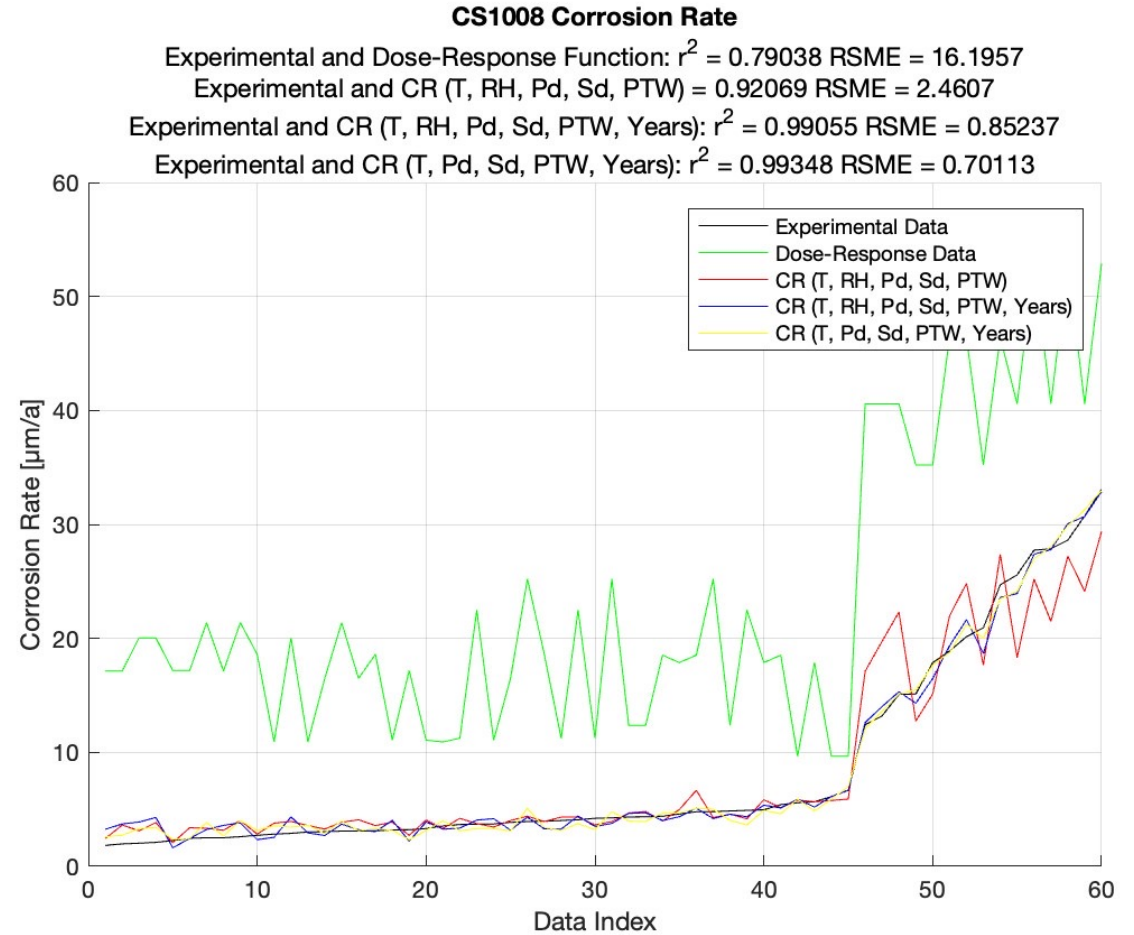
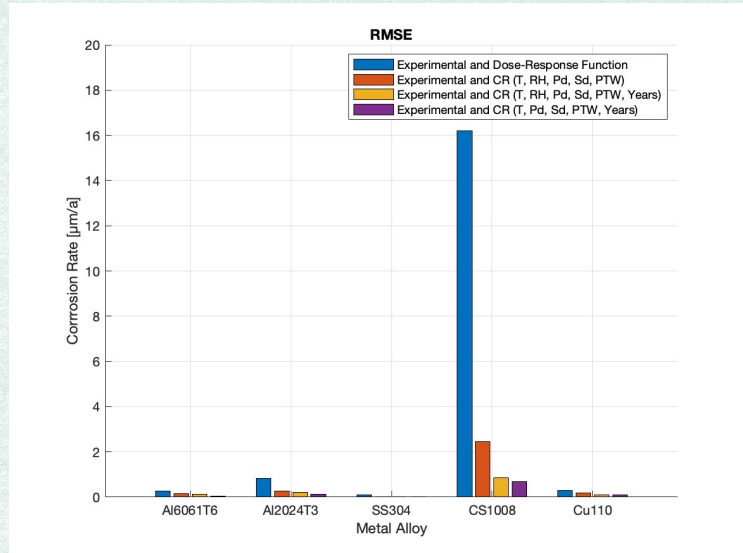
Results and Discussion

Corrosion Rate Estimation

$$r_{corr}(T, RH, Pd, Sd) = 1.77 \times P_d^{0.52} \times e^{0.02 \times RH + f_{st}} + 0.102 \times S_d^{0.62} \times e^{0.033 \times RH + 0.040 \times T}$$

$$f_{st} = 0.150 \times (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.054 \times (T - 10)$$

$$r_{corr}(T, Pd, Sd, PTW, \text{Years}) = 1.55 + (-0.113 - 0.416 \times T - 0.354 \times Pd + 0.598 \times Sd - 0.142 \times PTW + 0.044 \times T^2 + 0.009 \times Pd^2 - 0.005 \times Sd^2 + 0.0004 \times PTW^2 + 0.040(T \times Pd) - 0.048(T \times Sd) + 0.001(T \times PTW) + 0.0003 \times (Pd \times Sd) + 0.002(Pd \times PTW) + 0.009(Sd \times PTW)) \times \text{Years}^{-0.566}$$



Results and Discussion

Corrosivity Category

1008 Carbon Steel Corrosion Rate – 12 month Set 1 & 2							
Location	Exposure Angle [°]	Experimental		Dose-Response		Optimized Equation	
		Corrosion Rate [g/(m ² ×year)]	CC	Corrosion Rate [g/(m ² ×year)]	CC	Corrosion Rate [g/(m ² ×year)]	CC
Pacific Spaceport Complex – Alaska [Kodiak, AK]	0	194.42	C2	365.29	C3	186.08	C2
	0	164.76		277.19		157.09	
	30	158.63		365.29		168.46	
	30	140.69		277.19		138.69	
	45	148.72		365.29		148.94	
	45	119.04		277.19		121.71	
Port of Alaska [Anchorage, AK]	0	24.35	C2	168.27	C2	31.31	C2
	0	31.18		146.43		27.12	
	30	20.48		168.27		31.53	
	30	24.52		146.43		26.50	
	45	19.77		168.27		30.71	
	45	21.46		146.43		25.77	
University of Alaska Fairbanks [Fairbanks, AK]	0	33.21	C2	88.52	C2	24.46	C2
	0	27.90		85.94		33.48	
	30	28.91		88.52		24.32	
	30	23.73		85.94		30.47	
	45	31.69		88.52		24.62	
	45	22.28		85.94		30.47	
University of Alaska Anchorage [Anchorage, AK]	0	25.28	C2	135.29	C2	19.68	C2
	0	30.40		129.85		25.99	
	30	19.41		135.29		20.04	
	30	24.44		129.85		26.15	
	45	17.82		135.29		20.82	
	45	24.02		129.85		24.50	

Corrosivity Category	Corrosivity
C1	Very low
C2	Low
C3	Medium
C4	High
C5	Very high
CX	Extreme



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Thank You!
Questions?
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